

COURIER MANAGEMENT SYSTEM

Salesforce Administrator with AI AgentForce — A&S;

Project Documentation & Implementation Report

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Platform	Salesforce Lightning / Developer Edition
Project Type	Cloud-based Courier & Logistics Management System

Prepared as a consolidated project record with implementation evidence from the Salesforce org.

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Evidence note: The screenshots in this version are actual Salesforce screenshots supplied for this documentation. They are presented as implementation evidence; they are not generated mockups.

1. Introduction

Courier and logistics operations involve customers, shipments, delivery agents, branches, delivery updates and invoices. When these activities are handled through disconnected or manual processes, tracking and operational visibility become difficult. The Courier Management System (CMS) centralizes the core workflow on Salesforce.

The project uses Salesforce custom objects, relationships, validation rules, automation, Lightning navigation, role hierarchy, reports and dashboards. The objective is to provide a structured cloud application for recording shipment information, assigning delivery agents, updating delivery status and maintaining invoice details.

The implementation is designed as a practical academic project demonstrating Salesforce Administrator skills such as object creation, field configuration, relationships, validation, Flow automation, app configuration, security and reporting.

1.1 Objectives

- Create a centralized Salesforce application for courier operations.
- Maintain customer and shipment information in structured custom objects.
- Connect shipments with customers and delivery agents through relationships.
- Control data quality with validation rules.
- Automatically synchronize shipment status from delivery updates.
- Provide role-based access concepts and management visibility.
- Present operational information using reports and dashboards.

2. Ideation and Problem Statement

The project was identified from common courier-management challenges: manual shipment records, limited real-time visibility, difficulty maintaining delivery-agent information, inconsistent status updates and limited management reporting.

The proposed solution is a Salesforce-based CMS in which each major business entity is represented by a dedicated object. Related records provide a connected data model, while validation rules and Flow automation reduce inconsistent data entry.

2.1 Core workflow

- Customer information is registered and maintained.
- A shipment is created with source, destination, weight, price, expected date and status.
- A delivery agent can be associated with the shipment.
- Delivery updates record movement such as In Transit, Out for Delivery, Delivered or Delayed.
- A record-triggered Flow updates the related Shipment status from the latest Delivery Update.
- Invoices are associated with shipments and store amount, date and payment status.
- Reports and dashboards provide management-level visibility.

3. Requirement Analysis

The system requirements were divided into functional requirements and non-functional requirements.

3.1 Functional requirements

- Create, edit and view customer records.
- Create and track shipments using an auto-generated tracking number.
- Maintain delivery-agent and branch information.
- Record delivery updates linked to shipments.
- Create invoice records linked to shipments.
- Validate critical fields before records are saved.
- Automate shipment status changes from delivery updates.
- Provide application tabs, reports and dashboards.

3.2 Non-functional requirements

- Cloud accessibility through Salesforce.
- Structured and maintainable data model.
- Data-quality controls through validation rules.
- Role-based access design.
- Clear user navigation through a dedicated Lightning application.
- Usable reporting and dashboard presentation.

4. Project Design

The CMS data model uses six custom objects. Lookup relationships are used where records remain independently managed, while Delivery Update uses a Master-Detail relationship with Shipment.

Object	Purpose	Key Relationship
Customer__c	Stores customer contact and address information.	Parent/reference for Shipment
Shipment__c	Stores shipment tracking and delivery information.	Lookup → Customer; Lookup → Delivery Agent
Delivery_Agent__c	Stores agent identity, contact and branch.	Lookup → Branch
Branch__c	Stores branch location and manager information.	Referenced by Delivery Agent
Delivery_Update__c	Stores status updates for shipments.	Master-Detail → Shipment
Invoice__c	Stores invoice and payment information.	Lookup → Shipment

4.1 Relationship summary

Child / Related Object	Relationship	Parent / Referenced Object
Shipment__c	Lookup	Customer__c
Shipment__c	Lookup	Delivery_Agent__c
Delivery_Agent__c	Lookup	Branch__c
Delivery_Update__c	Master-Detail	Shipment__c
Invoice__c	Lookup	Shipment__c

5. Planning and Development

Development was organized around Salesforce configuration milestones: object creation, field creation, relationships, page layouts, validation rules, automation, app navigation, security, reports and dashboards.

5.1 Development sequence

Step	Implementation Activity
1	Create the Salesforce objects required for the courier workflow.
2	Configure standard and custom fields with suitable data types.
3	Configure lookup and master-detail relationships.
4	Arrange page layouts and application navigation.
5	Create validation rules for important data-quality checks.
6	Create a record-triggered Flow for shipment status synchronization.
7	Configure profiles and role hierarchy.
8	Create reports and dashboard visualizations.
9	Perform functional testing and review the application.

6. Salesforce Object & Field Configuration

The supplied Salesforce screenshots provide direct evidence of the object and field configuration. The following records were visible in the screenshots.

Object	Evidence visible in Salesforce
Customer	Address, City, Customer Name, Customer Type, Email, Is Active, Phone, Pincode, State and system fields.
Delivery Agent	Branch lookup, Delivery Agent Name, Email, Joining Date, Phone, Status and system fields.
Delivery Update	Remarks, Shipment master-detail, Status, Update Date, Update Number and system fields.
Branch	Branch Code, Branch Name, City, Manager Name, State and system fields.
Invoice	Amount, Invoice Date, Invoice Number, Payment Status, Shipment lookup and system fields.

The screenshots also show Salesforce Setup → Object Manager and the custom-tab configuration area used during the implementation.

7. Validation Rules

Validation rules were used to prevent invalid or incomplete data from being saved. The actual Salesforce evidence shows active rules on Customer, Shipment, Delivery Agent, Branch and Delivery Update.

Object	Rule	Purpose / Error Message
Customer	Email_Required	Email is required.
Customer	Validate_Phone_Number	Phone number must be 10 digits.
Customer	Validate_Pincode	Pincode must be 6 digits.
Shipment	Validate_Weight	Weight must be greater than zero.
Shipment	Validate_Price	Price must be greater than zero.
Shipment	Validate_Expected_Date	Expected delivery date cannot be in the past.
Delivery Agent	Validate_Agent_Email	Enter a valid email address.
Branch	Branch_Code_Required	Branch Code is mandatory.
Delivery Update	Remarks_Required_For_Delay	Remarks are required when status is Delayed.

8. Automation Flow

The project includes a record-triggered Flow named **Auto_Update_Shipment_Status**. It is designed to run when a Delivery Update record is created.

Flow logic

- Trigger: Delivery_Update__c record is created.
- Read the Shipment reference from \$Record.Shipment__c.
- Locate the related Shipment__c record using its Id.
- Set Shipment__c.Status__c equal to \$Record.Status__c.
- Save the updated Shipment record.

Trigger	Record Created
Source	Delivery_Update__c
Related record	Shipment__c
Field synchronized	Status__c
Flow label	Auto_Update_Shipment_Status

9. Lightning App and Navigation

A dedicated Lightning application titled **Courier Management System** was configured. The supplied screenshot shows the navigation bar containing the project modules.

Navigation tabs visible in the evidence include Customers, Shipments, Delivery Agents, Branches, Delivery Updates, Invoices, Reports and Dashboards.

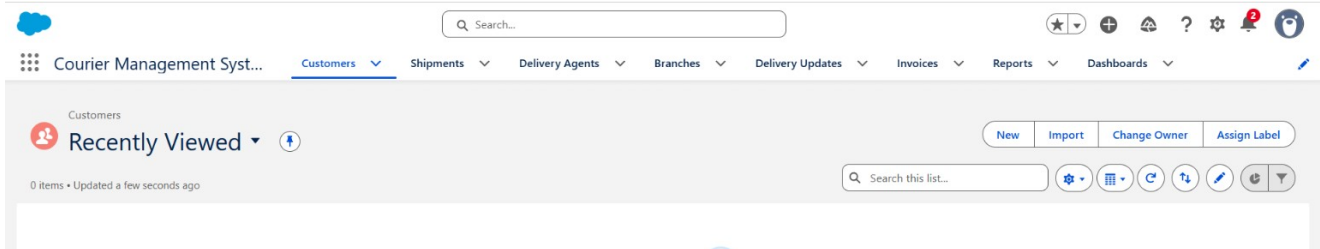


Figure 1. Salesforce Lightning application navigation for the Courier Management System.

10. Security and Role Structure

The project security design separates administrative, branch-management, delivery and customer-support responsibilities.

Role	Responsibility
CEO	Top-level organizational oversight.
Courier Admin	Application administration, configuration and operational management.
Branch Manager	Branch-level management and delivery operations.
Delivery Agent	Delivery execution and status updates.
Customer Support	Customer-facing support and shipment information assistance.

The hierarchy is designed as: CEO → Courier Admin → Branch Manager → Delivery Agent / Customer Support.

11. Reports and Dashboard

The reporting layer is intended to summarize operational performance without requiring users to inspect individual records.

Report	Purpose
Shipment Status Summary	Shows shipment distribution by delivery status.
Agent Performance Report	Provides delivery-agent activity/performance information.
Revenue by Branch Report	Summarizes invoice revenue by branch.

Dashboard components planned/configured for the project include Shipment Status Overview (donut), Agent Performance (bar) and Revenue by Branch (column).

12. Functional & Performance Testing

Testing focused on record creation, validation behavior, relationships, automation and navigation. The test cases below represent the intended verification matrix for the configured Salesforce solution.

Test Case	Expected Result
Create Customer with missing email	Email validation prevents save.
Enter invalid phone length	Phone validation prevents save.
Enter invalid pincode length	Pincode validation prevents save.
Create Shipment with weight ≤ 0	Weight validation prevents save.
Create Shipment with price ≤ 0	Price validation prevents save.
Use past Expected Date	Expected-date validation prevents save.
Create Delivery Agent with invalid email	Agent email validation prevents save.
Create Branch without Branch Code	Branch code validation prevents save.
Create Delayed Delivery Update without remarks	Remarks validation prevents save.
Create valid Delivery Update	Related Shipment status is synchronized by Flow.
Open CMS app	Configured navigation tabs are available.

13. Advantages and Limitations

13.1 Advantages

- Centralized cloud-based courier data.
- Structured relationships between business records.
- Validation rules improve data consistency.
- Flow automation reduces repetitive status updates.
- Lightning navigation provides a single application entry point.
- Reports and dashboards support operational visibility.

13.2 Limitations

- The academic implementation depends on Salesforce configuration and available platform features.
- External GPS, SMS and mobile integrations are not part of the demonstrated core configuration.
- Operational performance depends on correct data entry and appropriate user permissions.

14. Conclusion

The Courier Management System demonstrates how Salesforce can be configured as a connected courier and logistics application. The implementation covers the core data model, relationships, validation rules, automation, application navigation, role structure and reporting design.

The supplied Salesforce evidence confirms the configuration of multiple custom objects, fields and active validation rules, along with the Lightning application navigation. Together, these components provide a coherent foundation for managing customers, shipments, delivery operations and invoices.

15. Future Scope

The project can be extended beyond the current academic scope with additional integrations and automation.

- Mobile tracking application for customers and delivery agents.
- GPS/location integration for live delivery tracking.
- SMS or WhatsApp notifications for shipment milestones.
- AI-based delivery-route or delivery-time optimization.
- Customer self-service tracking portal.
- Advanced analytics for branch, agent and delivery performance.
- Integration with external payment and invoicing services.

16. Appendix — Salesforce Implementation Evidence

The following figures are the actual screenshots supplied for the project documentation. They are grouped by the Salesforce configuration area they demonstrate.

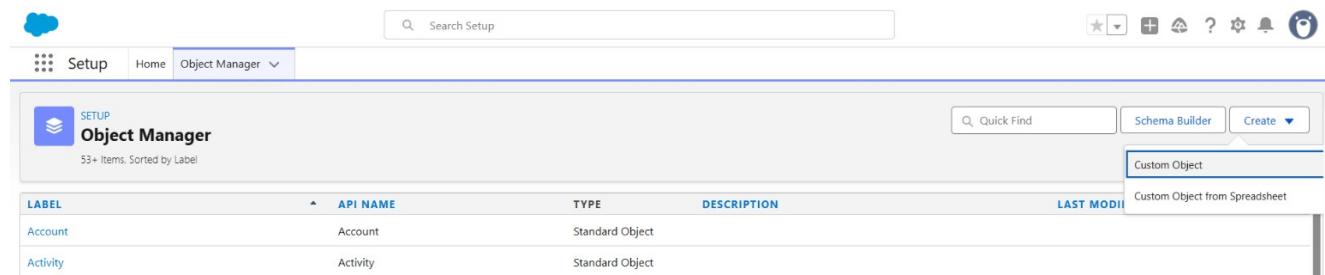
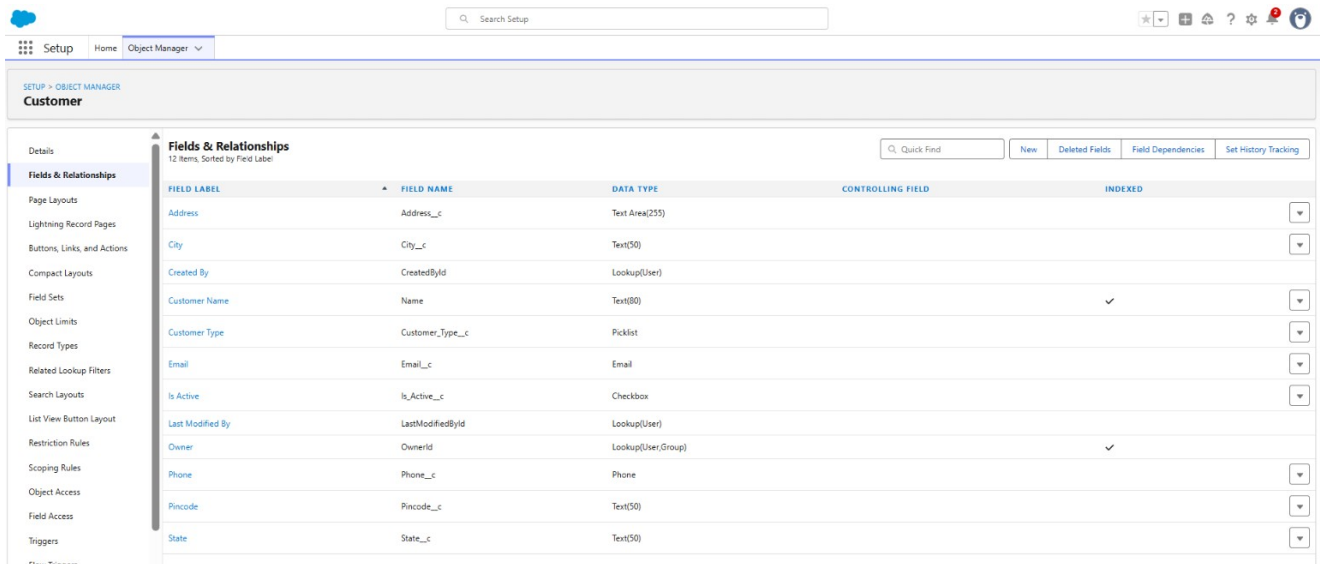


Figure 2. Salesforce Object Manager showing the custom-object creation option.

The screenshot displays the Salesforce Setup interface. At the top, there's a search bar labeled 'Search Setup'. Below it, the 'Setup' menu is visible with 'Home' and 'Object Manager' tabs. The left sidebar shows 'User Interface' expanded, with 'Rename Tabs and Labels' and 'Tabs' options. The main content area is titled 'Custom Tabs' and includes a description of custom tabs, a 'Custom Object Tabs' section with a 'New' button and a 'What Is This?' link, and a message stating 'No Custom Object Tabs have been defined'.

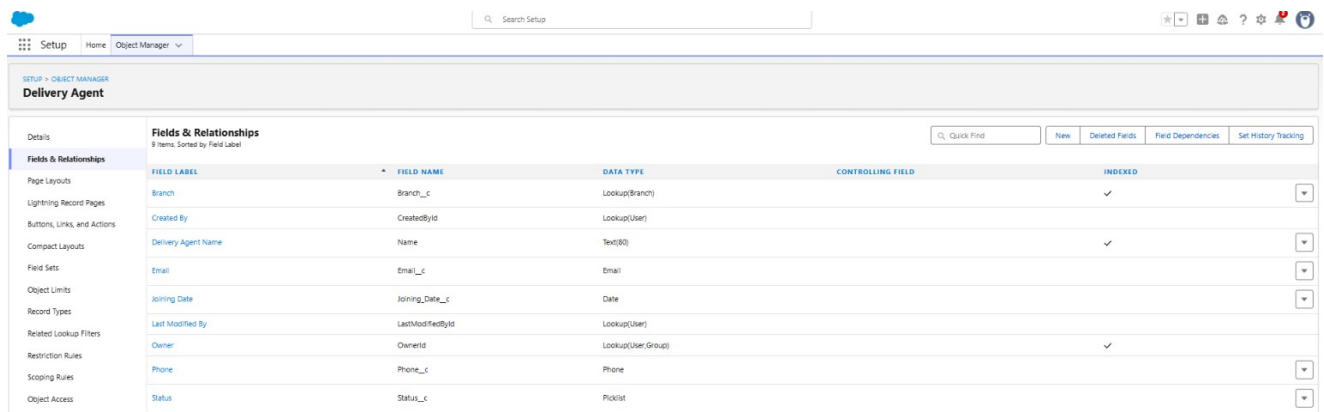
Figure 3. Salesforce Setup → Tabs page showing the Custom Object Tabs section.



The screenshot shows the Salesforce Setup interface for the 'Customer' object. The 'Fields & Relationships' section is active, displaying a table of 12 fields. The table has columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The fields are sorted by Field Label.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text Area(255)		
City	City__c	Text(50)		
Created By	CreatedById	Lookup(User)		
Customer Name	Name	Text(80)		✓
Customer Type	Customer_Type__c	Picklist		
Email	Email__c	Email		
Is Active	Is_Active__c	Checkbox		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		
Pincode	Pincode__c	Text(50)		
State	State__c	Text(50)		

Figure 4. Customer object Fields & Relationships showing configured fields and data types.

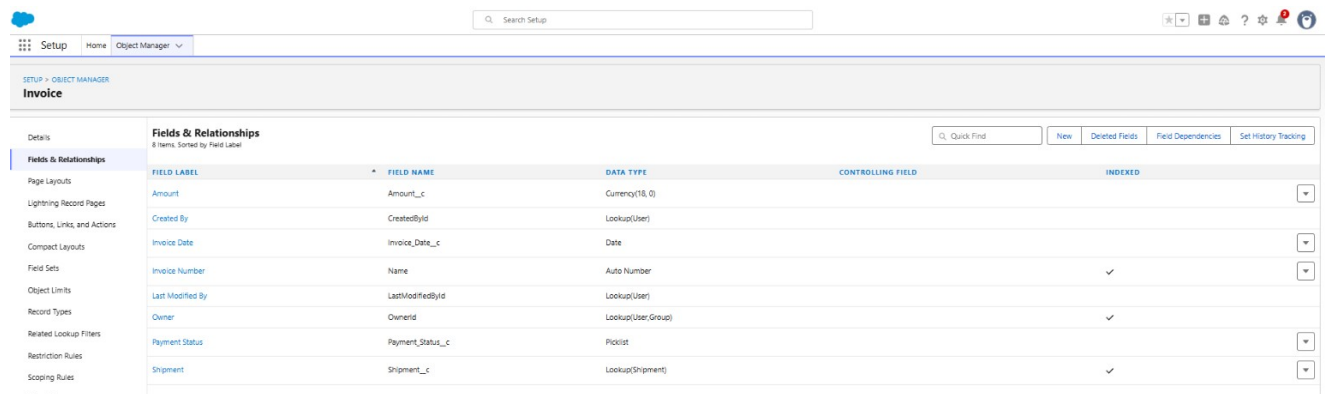


FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Branch	Branch__c	Lookup(Branch)		✓
Created By	CreatedById	Lookup(User)		
Delivery Agent Name	Name	Text(80)		✓
Email	Email__c	Email		
Joining Date	Joining_Date__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		
Status	Status__c	Picklist		

Figure 5. Delivery Agent object Fields & Relationships showing Branch lookup and agent fields.

Branch					
Fields & Relationships					
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED	
Branch Code	Branch_Code__c	Text(255)			
Branch Name	Branch_Name__c	Text(255)			
Branch Name	Name	Text(80)		✓	
City	City__c	Text(255)			
Created By	CreatedById	Lookup(User)			
Last Modified By	LastModifiedById	Lookup(User)			
Manager Name	Manager_Name__c	Text(255)			
Owner	OwnerId	Lookup(User, Group)		✓	
State	State__c	Text(255)			

Figure 6. Branch object Fields & Relationships showing branch information fields.



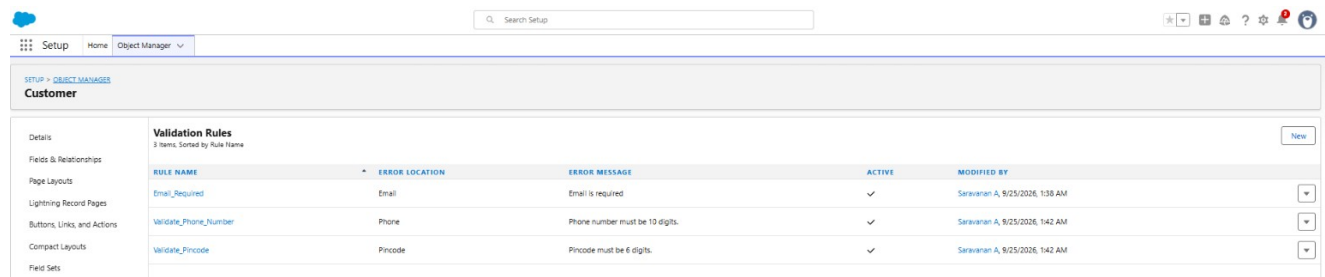
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Amount	Amount__c	Currency(16, 0)		
Created By	CreatedById	Lookup(User)		
Invoice Date	Invoice_Date__c	Date		
Invoice Number	Name	Auto Number		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User, Group)		✓
Payment Status	Payment_Status__c	Picklist		
Shipment	Shipment__c	Lookup(Shipment)		✓

Figure 7. Invoice object Fields & Relationships showing amount, invoice date, payment status and shipment lookup.

The screenshot shows the Salesforce Setup interface for the 'Delivery Update' object. The left sidebar contains navigation links: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, and Restriction Rules. The main content area is titled 'Fields & Relationships' and contains a table with 7 items, sorted by Field Label. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The rows are: Created By (CreatedById, Lookup(User)), Last Modified By (LastModifiedById, Lookup(User)), Remarks (Remarks__c, Text(255)), Shipment (Shipment__c, Master-Detail (Shipment)), Status (Status__c, Picklist), Update Date (Update_Date__c, DateTime), and Update Number (Update_Number, Auto Number). The 'Shipment' row is highlighted, indicating a master-detail relationship. The 'Status' row is also highlighted, indicating a picklist field. The 'Update Date' row is highlighted, indicating a date-time field. The 'Update Number' row is highlighted, indicating an auto-number field.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Remarks	Remarks__c	Text(255)		
Shipment	Shipment__c	Master-Detail (Shipment)		✓
Status	Status__c	Picklist		
Update Date	Update_Date__c	DateTime		
Update Number	Update_Number	Auto Number		✓

Figure 8. Delivery Update object Fields & Relationships showing Shipment master-detail, Status and Update Date.



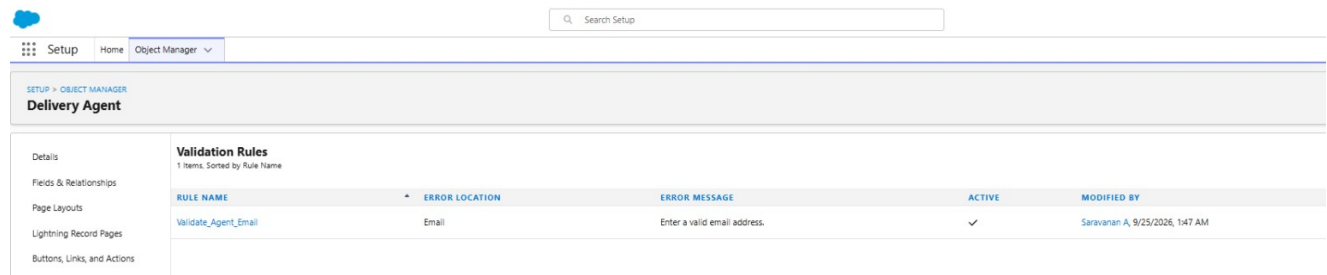
The screenshot shows the Salesforce Setup interface for the 'Customer' object. The left sidebar contains navigation links: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, and Field Sets. The main content area is titled 'Validation Rules' and shows 3 items. A table lists three active validation rules: 'Email_Required' for the Email field, 'Validate_Phone_Number' for the Phone field, and 'Validate_Pincode' for the Pincode field. All rules are active and were last modified by Saravanan A. on 9/25/2026.

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Email_Required	Email	Email is required	✓	Saravanan A. 9/25/2026, 1:38 AM
Validate_Phone_Number	Phone	Phone number must be 10 digits.	✓	Saravanan A. 9/25/2026, 1:42 AM
Validate_Pincode	Pincode	Pincode must be 6 digits.	✓	Saravanan A. 9/25/2026, 1:42 AM

Figure 9. Customer validation rules showing active email, phone and pincode checks.

Validation Rules				
RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Validate_Expected_Date	Expected Date	Expected delivery date cannot be in the past.	✓	Saravanan A. 9/25/2026, 1:46 AM
Validate_Price	Price	Price must be greater than zero.	✓	Saravanan A. 9/25/2026, 1:45 AM
Validate_Weight	Weight	Weight must be greater than zero.	✓	Saravanan A. 9/25/2026, 1:44 AM

Figure 10. Shipment validation rules showing active expected-date, price and weight checks.



The screenshot shows the Salesforce Setup interface. At the top, there is a search bar labeled "Search Setup". Below it, the navigation menu includes "Setup", "Home", and "Object Manager". The main content area is titled "Delivery Agent" and "Validation Rules". A sidebar on the left lists navigation options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, and Buttons, Links, and Actions. The main table displays one validation rule named "Validate_Agent_Email".

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Validate_Agent_Email	Email	Enter a valid email address.	✓	Saravanan A , 9/25/2026, 1:47 AM

Figure 11. Delivery Agent validation rule showing the active email validation.

The screenshot shows the Salesforce Setup interface. At the top, there's a navigation bar with 'Setup', 'Home', and 'Object Manager'. Below this, the breadcrumb trail reads 'SETUP > OBJECT MANAGER > Branch'. The main content area is titled 'Validation Rules' and shows a list of rules for the 'Branch' object. A single rule, 'Branch_Code_Required', is listed as active. The left sidebar contains navigation links for 'Details', 'Fields & Relationships', 'Page Layouts', 'Lightning Record Pages', and 'Buttons, Links, and Actions'.

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Branch_Code_Required	Branch Code	Branch Code is mandatory.	✓	Saravanan A , 5/25/2026, 1:54 AM

Figure 12. Branch validation rule showing the active Branch Code requirement.

Setup Home Object Manager

SETUP > OBJECT MANAGER

Delivery Update

Validation Rules

1 items. Sorted by Rule Name

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Remarks_Required_For_Delay	Remarks	Remarks are required when status is Delayed.	✓	Saravanan A, 9/25/2020, 1:56 AM

Figure 13. Delivery Update validation rule showing the active delayed-status remarks requirement.

Project Evidence Summary

The evidence set supplied for this version demonstrates Salesforce Setup, Object Manager, custom-object tabs, field configurations, Lightning application navigation and active validation rules across the main CMS objects.

Where a feature is described in the report but no direct screenshot was supplied in this batch, the description is based on the project configuration already documented rather than presented as screenshot evidence.

End of Project Documentation